

Vedansh Kokra

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EXPERIENCE

AI/ML Engineer

Sept. 2025 – Apr. 2026

Compass Data and Intelligence (4 Month Full-time Co-op, 4 Month Contract Part-time)

Toronto, ON

- Engineered production-grade multi-agent AI systems with parallel tool calling, dynamic prompt injection, and content moderation guardrails, designing modular orchestration workflows and services for scalable enterprise AI applications.
- Developed a Snowflake-native multi-agent system leveraging Cortex Search and Cortex Agents to reason over distributed enterprise data, enabling semantic retrieval and coordinated agent execution across fragmented data sources.
- Refactored production AI infrastructure to adopt the Model Context Protocol (MCP) architecture, building a dynamic prompt management layer with runtime context injection alongside reusable microservices and standardized interfaces that improved interoperability, extensibility, and deployment scalability.
- Built a standardized LLM evaluation framework with LLM-as-judge scoring, tool invocation pattern analysis, automated benchmarking, and latency monitoring, templated for plug-and-play adoption across Compass AI agents through environment-based configuration.
- Conducted structured comparative evaluation of LLM observability platforms including Braintrust, LangSmith, Arize, and first-party solutions from OpenAI and Google, establishing standardized evaluation criteria and informing internal platform adoption decisions.

AI/ML Engineer

May 2025 – Aug. 2025

Carta Maple Technologies (4 Month Full-time Co-op)

Kitchener, ON

- Architected an AI-powered rule recommendation engine for financial transaction reconciliation across millions of transactions, leveraging semantic retrieval and parameter-level pattern extraction to generate context-grounded candidate rules via LLMs, achieving over 90% recommendation accuracy.
- Engineered a reusable semantics-based rule overlap detection system using subset/superset relationship analysis, replacing brute-force transaction scanning with rule-level reasoning to enable intelligent rule lifecycle management across millions of financial records.
- Extended and optimized a large-scale monolithic backend by developing standardized, reusable service abstractions that improved the performance and maintainability of core reconciliation workflows across the platform.

AI/ML Engineer

Sept. 2024 – Dec. 2024

Guidepoint Global LLC (4 Month Full-time Co-op)

New York, USA

- Architected an end-to-end LLM evaluation infrastructure bridging development and production, implementing automated pre-deployment benchmarking against golden datasets with comparative scoring against the live model and enforcing quality gates to block regressions.
- Extended the evaluation platform into production with continuous real-time assessment on live data and human-in-the-loop feedback loops for dataset curation and quality validation.
- Engineered a CI/CD pipeline with MLflow Model Registry and Delta Live Tables, integrating evaluation into automated deployment workflows with real-time monitoring of latency, throughput, and response quality metrics.
- Developed a RAG evaluation system with custom LLM judges measuring retrieval precision and semantic relevance while implementing automated retraining triggers based on statistical drift detection and continuous performance assessment.

AI Software Developer

Jan. 2024 – Apr. 2024

Locobuzz Ltd. (4 Month Full-time Co-op)

Mumbai, India

- Spearheaded AI solution projects including Chat with Data (natural language interface for data queries) and Ticket Categorization (intelligent ticket routing) leveraging open-source Large Language Models.
- Designed and deployed a scalable microservices infrastructure using FastAPI and Docker, enabling seamless integration with the existing .NET ecosystem.
- Engineered production-grade RAG pipelines using LangChain, incorporating chunk optimization, hybrid search, and custom prompting strategies to enhance retrieval accuracy and response quality.
- Developed a fine-tuning pipeline integrating PEFT techniques and quantization alongside automated evaluation workflows in LangSmith for tracking model performance and deployment readiness.

TECHNICAL SKILLS

AI/ML & Agentic Systems: LLM Orchestration & Multi-Agent Systems, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), LLM-as-Judge Evaluation, Fine-Tuning (PEFT, Quantization), Prompt Engineering, LangChain, LangGraph, LangSmith, PyTorch, TensorFlow

MLOps & Infrastructure: MLflow (Model Registry), Delta Live Tables, Databricks, Snowflake (Cortex Search, Cortex Agents), CI/CD Pipelines, Model Evaluation & Observability (Braintrust, Arize)

Backend & APIs: Python, FastAPI, Docker, Microservices Architecture, SQL, Node.js

Languages: Python, C++, C#, Objective-C, R, Bash